

# POCKETSMART AI

Your Smart Budget & Recommendation Assistant

## PROJECT DOCUMENTATION

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## Introduction and Objectives

PocketSmart AI is a web-based Smart Budget and Recommendation Assistant designed to help users plan and manage their requirements within a defined budget. The platform brings budget planning and Generative AI together in a single authenticated web application.

It provides three specialized planning modules: Home Interior Budget Planner, Party Budget Planner, and Jewelry Budget Planner. Each module collects domain-specific information while following a common workflow: user input, validation, AI processing, structured results, and History. Users enter budgets, quantities, preferences, event or room details, and other requirements. The FastAPI backend validates the request and uses the Gemini API to generate personalized recommendations. Results include estimated costs, quantities, category-wise recommendations, budget summaries, shopping references, and additional suggestions.

### Project Objectives

- Provide personalized recommendations according to the user's available budget.
- Support Home Interior, Party Planning, and Jewelry Selection through dedicated modules.
- Collect structured planning information and present readable recommendation results.
- Protect application routes through authentication and maintain recommendation history for authenticated users.

## 2. Problem Statement and Scope

Choosing suitable products, services, or arrangements can be difficult when users have many options and must remain within a fixed budget. Home interiors may require furniture, lighting, fans, and dining tables; parties may require venue, catering, decoration, and entertainment; jewelry selection may depend on budget, occasion, style, and outfit.

PocketSmart AI addresses this problem by converting budget and planning requirements into a structured recommendation process. The system reduces manual comparison effort while keeping the final suggestions organized around the user's available budget.

| Module               | Function                                                |
|----------------------|---------------------------------------------------------|
| Registration / Login | Creates and authenticates user accounts                 |
| Dashboard            | Provides access to planning modules                     |
| Home Planner         | Collects home interior requirements                     |
| Party Planner        | Collects event planning requirements                    |
| Jewelry Planner      | Collects jewelry requirements and optional outfit image |
| Recommendations      | Generates structured, budget-aware suggestions          |

### 3. System Architecture and Technology Stack

PocketSmart AI follows a layered application architecture. The major flow is: User Input → Validation → FastAPI Route → Planner Logic → Gemini API → Result Formatting → Template Display → History.

#### Major Layers

| Layer                 | Responsibility                                                                         |
|-----------------------|----------------------------------------------------------------------------------------|
| Frontend / Templates  | Forms, navigation, result pages, cards, summaries, and user interface components.      |
| FastAPI Backend       | Routing, request processing, validation, authentication, and application coordination. |
| Planner Modules       | Process Home, Party, and Jewelry-specific requirements.                                |
| Gemini AI Integration | Generate personalized recommendation content from planning context.                    |
| History               | Store and display previous recommendation results for authenticated users.             |

#### Technology Stack

| Technology                  | Purpose                                   |
|-----------------------------|-------------------------------------------|
| Python                      | Backend programming and application logic |
| FastAPI                     | Web framework and routing                 |
| Jinja2                      | Server-rendered HTML templates            |
| HTML/CSS                    | Frontend structure and styling            |
| Gemini API                  | AI-based recommendation generation        |
| JWT / Cookie Authentication | Authenticated access                      |
| Pydantic                    | Data validation and modelling             |
| Pillow                      | Image handling                            |
| Uvicorn                     | ASGI application server                   |
| python-dotenv               | Environment configuration                 |

## 4. Project Structure, Backend and Authentication

The project separates application routes, authentication, Gemini utilities, data models, templates, and static resources.

### Project Structure

```
pocketsmar
t_code/
├──
pocketsmar
t/
├──
static/
├──
templates/
├──
.env
├──
app.py
├──
auth.py
├──
gemini_util
ls.py
├──
models.py
├──
README.md
├──
requiremen
ts.txt
└── venv/
```

### Backend Files

| File             | Role                                              |
|------------------|---------------------------------------------------|
| app.py           | Main FastAPI application and route orchestration. |
| auth.py          | Authenticatio n and token-related operations.     |
| gemini_utils.py  | Gemini API interaction utilities.                 |
| models.py        | Pydantic/data model definitions.                  |
| requirements.txt | Project dependency list.                          |

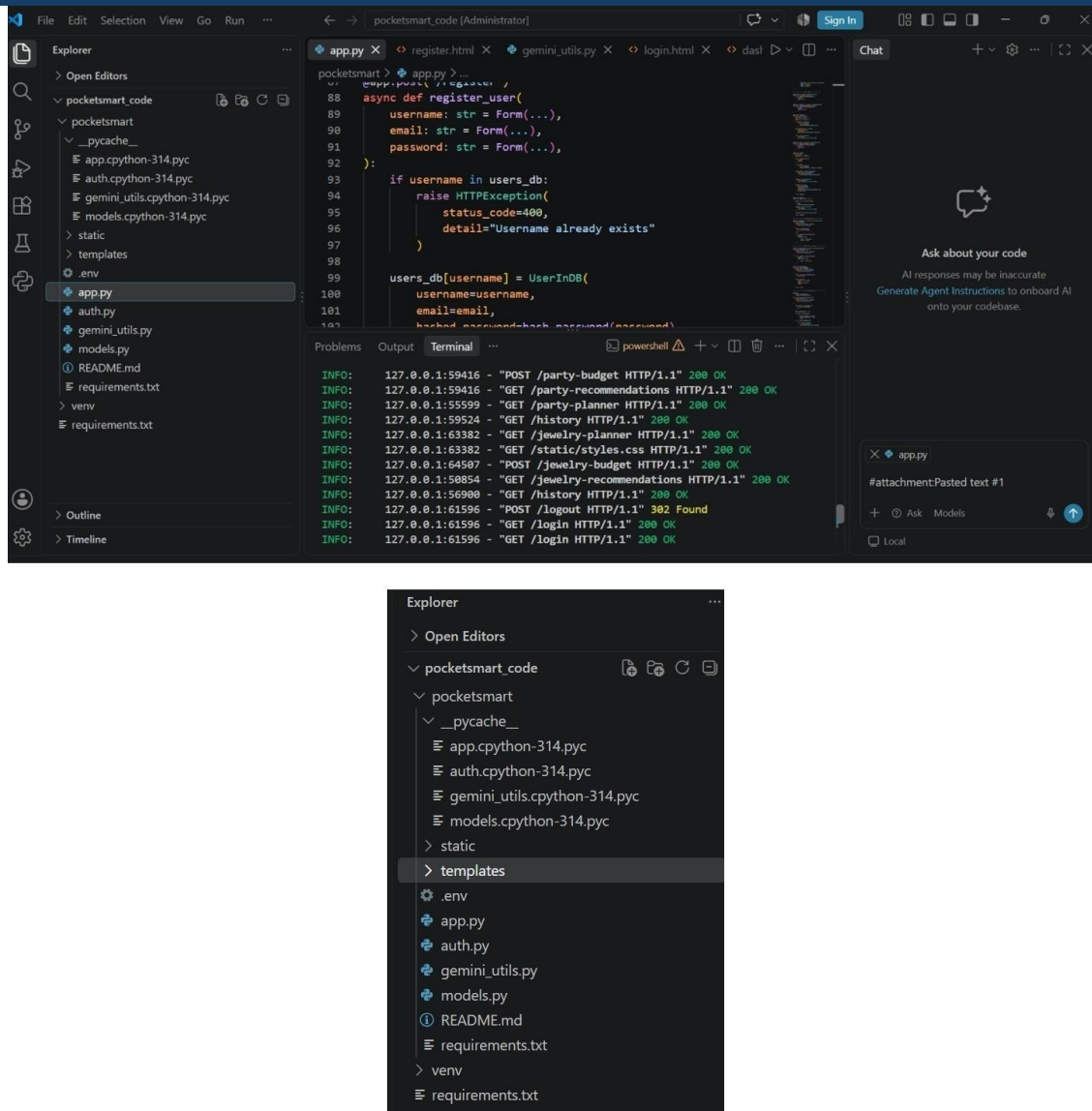


Figure 4.1 – Project File Explorer and expanded development view

Authentication Flow: Registration creates a user account after validation. Passwords are hashed before storage. During login, credentials are checked and a JWT/access token is issued in an HTTP-only cookie so protected routes can identify the authenticated user. Logout invalidates the active token/session.

The project explorer and development view together show the application directory, backend Python files, templates, static resources, environment configuration, dependency files, authentication logic, and route activity.

## 5. Gemini AI Integration

Gemini is a core component of PocketSmart AI. The application sends the selected planner's budget, preferences, quantities, and other requirements through the Gemini utility layer. Gemini generates context-aware recommendation content that the backend formats for the corresponding result page.

### How Gemini Is Used

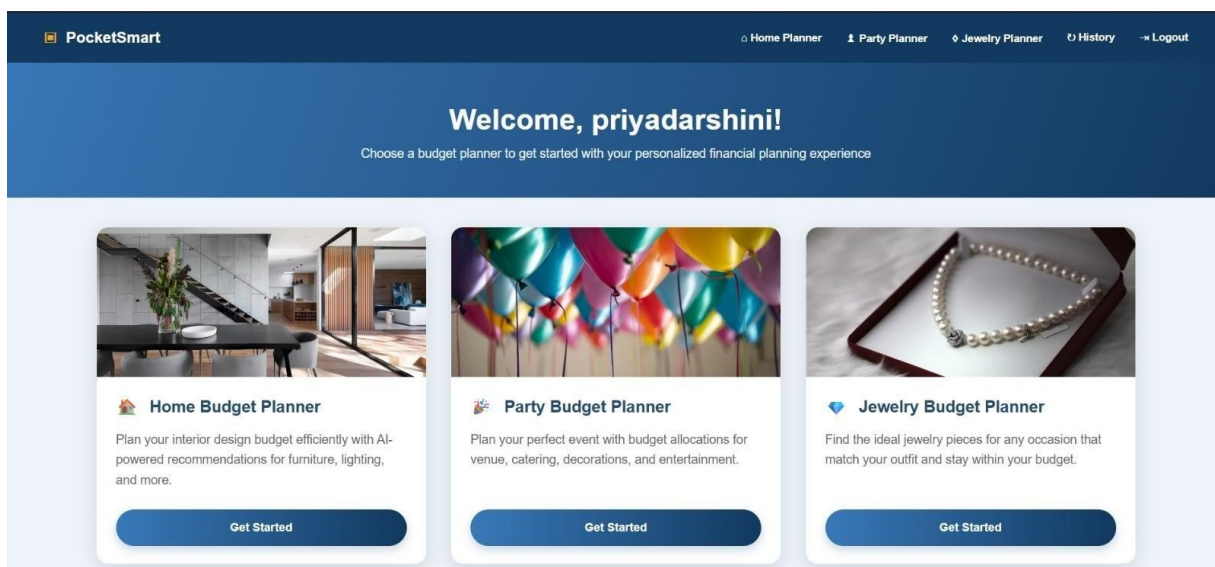
- Home planning: budget-aware interior recommendations based on rooms, quantities, furniture, lighting, and additional requirements.
- Party planning: recommendations based on budget, guest count, event type, venue type, and selected party needs.
- Jewelry planning: suggestions based on budget, occasion, style preferences, and optional outfit-image context.

The backend converts AI-generated content into structured categories containing item descriptions, estimated prices, quantities, budget summaries, and additional suggestions. The Gemini API key is loaded through environment configuration rather than being exposed in application pages.

## 6. Dashboard and Main Navigation

After authentication, the user reaches the PocketSmart AI dashboard. It provides access to the Home Budget Planner, Party Budget Planner, Jewelry Budget Planner, History, and Logout.

*Figure 6.1 – PocketSmart AI Dashboard*



## 7. Home Interior Budget Planner

The Home Interior Budget Planner helps users plan interior-related requirements according to a specified budget. It collects total budget, lights/fixtures, ceiling fans, furniture pieces, dining tables, rooms to include, and additional requirements or preferences.

Figure 7.1 – Home Interior Budget Planner

**Home Interior Budget Planner**  
Create a customized budget plan for your dream home interior

**Budget Details**  
Total Budget (₹)  
25000

**Fixtures & Furniture**  
Number of Lights/Fixtures: 8  
Number of Ceiling Fans: 6  
Number of Furniture Pieces: 4  
Number of Dining Tables: 2

**Rooms to Include**  
☒ Living Room ☒ Kitchen ☒ Bedroom

**Additional Information**  
Special Requirements or Preferences  
Any specific requirements or preferences...

**Generate Recommendations**

The user enters the budget, specifies required quantities, selects rooms, adds preferences, and selects Generate Recommendations. This keeps the request structured before it is sent to the recommendation service.

## 8. Home Recommendation Output

The Home output contains a Budget Summary and category-based recommendations for Lighting & Fixtures, Ceiling Fans, Dining Tables, and Furniture Pieces. Each recommendation can include an item name, description, estimated price, quantity, and shopping links. The result also includes category allocation and additional suggestions.

Home Interior Recommendations Page:  
Displays AI-generated product suggestions for home decor based on the user's budget and style preferences.

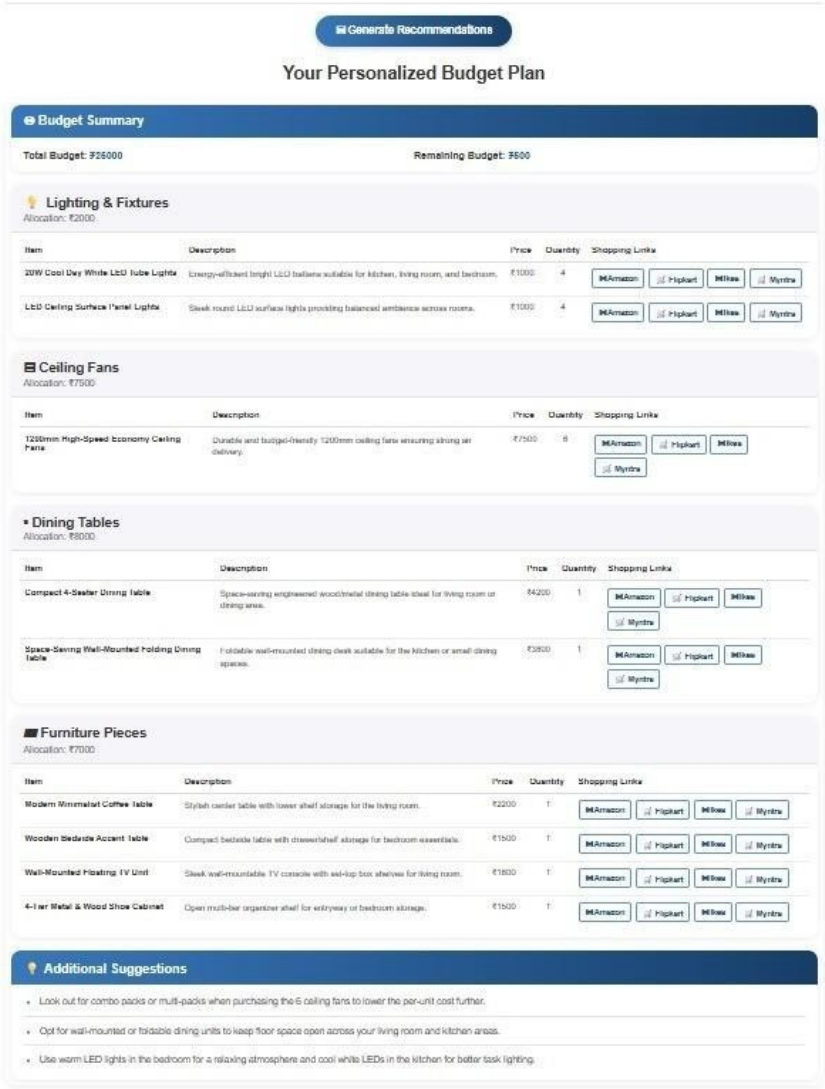


Figure 8.1 – Home Recommendation Output

## 9. Party Budget Planner

The Party Budget Planner is designed for event-specific budget planning. The user provides total budget, guest count, party type, venue type, party needs, and additional requirements. Party needs can include catering, decoration, and entertainment; the additional field supports themes, special requests, and dietary restrictions.

## Party Budget Planner

Plan your perfect event with AI-powered budget recommendations

### Basic Information

Total Budget (₹)

10000

Number of Guests

10

### Event Details

Party Type

Birthday

Venue Type

Outdoor

### Party Needs

☒ Catering

☒ Decoration

☒ Entertainment

### Additional Requirements

Special requests, themes, dietary restrictions, etc.

Generate Budget Plan

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Figure 9.1 – Party Budget Planner

## 10. Party Recommendation Output

The Party Recommendation page presents the generated event plan according to the user's requirements and budget. The result separates event costs into clear categories so the user can understand how the budget is distributed.



# Your Party Budget Plan

5 Preset Items

Budget: \$10000.00

## Venue

\$1000.00

### Public Park / Garden Space Permit

Permit or rental fee for reserving a designated place (e.g., a local park or society garden).

Quantity: 1

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

## Catering

\$4600.00

### 1.5 kg Customized Birthday Cake

Customized chocolate or fruit cake suitable for 10 guests.

Quantity: 1

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

### Snack & Main Course Combo (Biryani/Pizza)

Individual meal boxes with main course, starter, and soft drinks for 10 people.

Quantity: 10

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

### Beverages & Refreshments Pack

Assorted cold drinks, juice boxes, and packaged drinking water bottles.

Quantity: 1

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

## Decoration

\$2000.00

### Outdoor DIY Birthday Decoration Kit

Includes pastel balloons, metallic foil banner, battery-powered fairy lights, paper masks, and paper props.

Quantity: 1

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

## Entertainment

\$1600.00

### Portable Bluetooth Party Speaker Rental

High-bass battery-powered portable speaker suitable for outdoor music.

Quantity: 1

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

### Party Games Bundle

1 set card games (UNO, Pictionary/Charades deck) and outdoor games like Pictionary or Scattergories.

Quantity: 1

Step-on: [cancel](#) [ticket](#) [invite](#) [notes](#) [bookmyshow](#)

Total Budget:

\$10000.00

Allocated:

\$0.00

Remaining:

\$10000.00

## Additional Suggestions

- Ask guests to bring outdoor picnic mats or folding chairs for extra comfort.
- Prepare a collaborative Spotify playlist before the event so guests can add their favorite songs.
- Carry eco-friendly garbage bags to ensure quick cleanup of the outdoor area after the party.

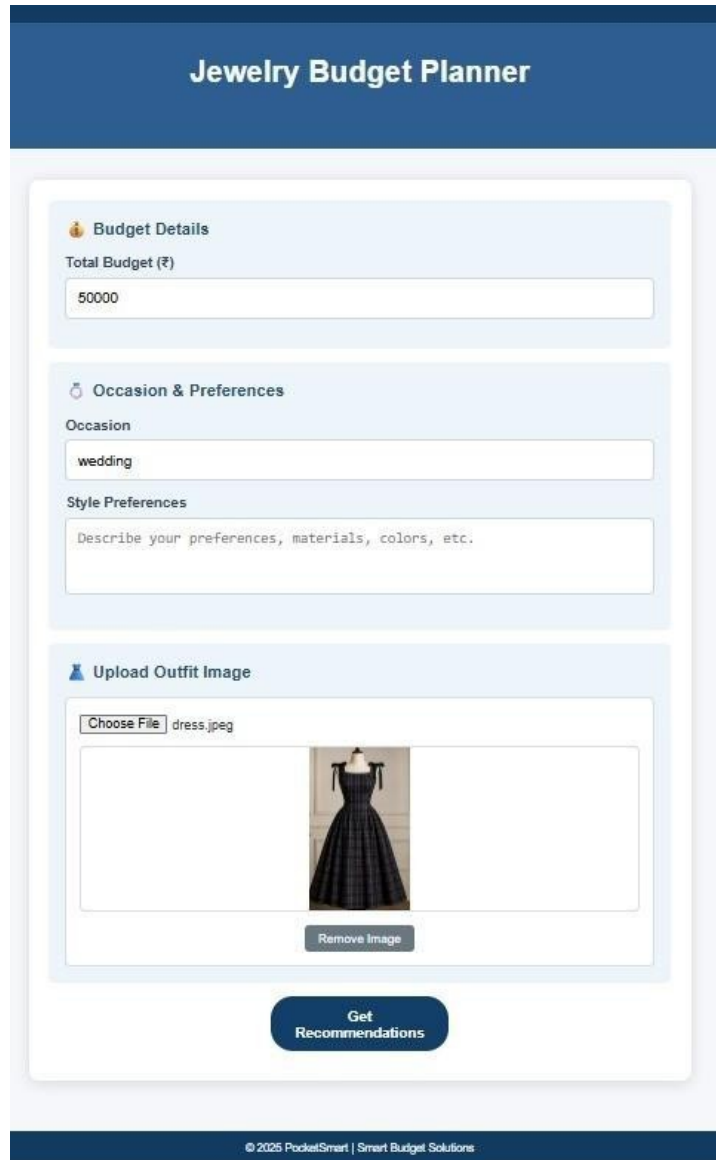
[← Back to Party Planner](#)

Figure 10.1 – Party Recommendation Output

| Category               | Output                                                 |
|------------------------|--------------------------------------------------------|
| Venue                  | Allocation, quantity, and estimated cost.              |
| Catering               | Food and beverage recommendations and allocation.      |
| Decoration             | Event appearance and setup recommendations.            |
| Entertainment          | Equipment, activities, or entertainment suggestions.   |
| Budget Summary         | Total Budget, Allocated Amount, and Remaining Balance. |
| Additional Suggestions | Further event-planning guidance.                       |

## 11. Jewelry Budget Planner

The Jewelry Budget Planner extends the common workflow with occasion, style preferences, and optional outfit-image information. The user provides total budget, occasion, style preferences, and an optional outfit image for additional visual context.



The screenshot displays the 'Jewelry Budget Planner' interface. It features a dark blue header with the title 'Jewelry Budget Planner'. Below the header, the form is organized into three main sections: 'Budget Details', 'Occasion & Preferences', and 'Upload Outfit Image'. The 'Budget Details' section includes a 'Total Budget (₹)' input field with the value '50000'. The 'Occasion & Preferences' section contains an 'Occasion' input field with the value 'wedding' and a 'Style Preferences' text area with the placeholder text 'Describe your preferences, materials, colors, etc.'. The 'Upload Outfit Image' section includes a 'Choose File' button, a file name 'dress.jpeg', a preview image of a black and white dress, and a 'Remove Image' button. At the bottom of the form is a large blue button labeled 'Get Recommendations'. The footer of the page shows the copyright notice '© 2025 PocketSmart | Smart Budget Solutions'.

Figure 11.1 – Jewelry Budget Planner

## 12. Jewelry Recommendation Output

The Jewelry Recommendation page presents personalized suggestions based on the submitted budget, occasion, style preferences, and optional outfit information. It combines structured text inputs with optional visual context from the outfit image.

**Jewelry Budget Recommendations Page:** Presents personalized jewelry options tailored to the occasion, outfit, and budget, using Gemini-generated insights.

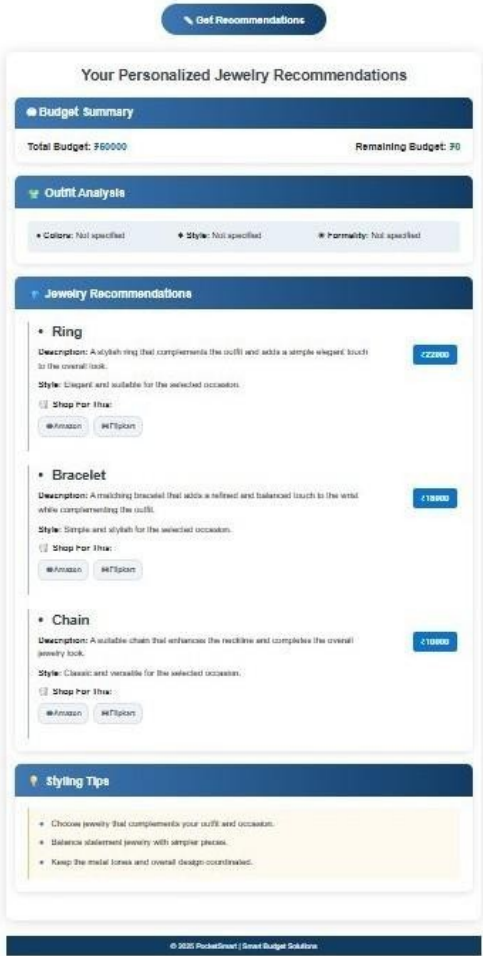
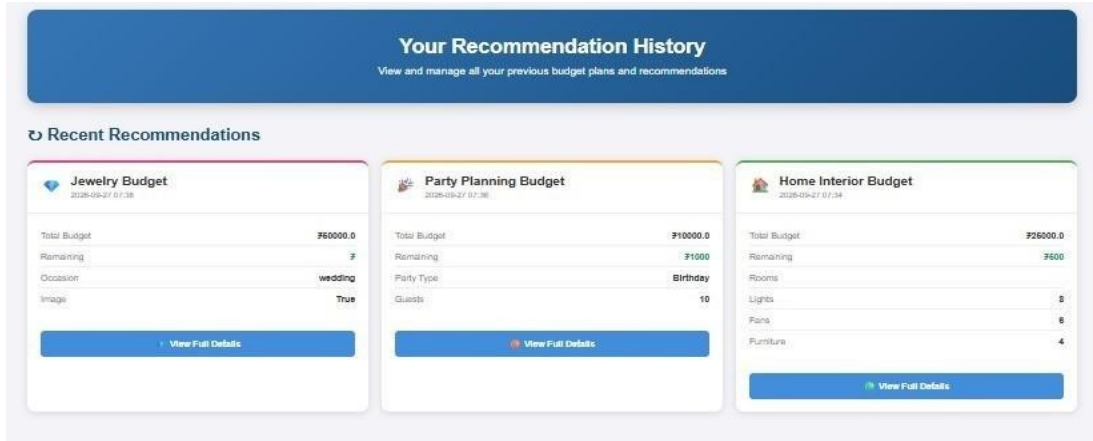


Figure 12.1 – Jewelry Recommendation Output

| Component               | Details                                                                              |
|-------------------------|--------------------------------------------------------------------------------------|
| Budget Summary          | Total budget and remaining budget.                                                   |
| Outfit Analysis         | Information derived from the submitted outfit context.                               |
| Jewelry Recommendations | Ring, Bracelet, Chain, description, style, estimated price, and shopping references. |
| Styling Tips            | Guidance for coordinating jewelry with the outfit and occasion.                      |

## 13. Recommendation History

PocketSmart AI provides a History module for authenticated users. Records can include Jewelry Budget, Party Planning Budget, and Home Interior Budget entries with total budget, remaining budget, planning-s



pecific details, date/time, image availability where applicable, and a View Full Details option.

Figure 13.1 – Recommendation History

## 14. Frontend Templates and Application Routes

Jinja2 templates render the user-facing pages.

| Template / Route                                        | Purpose                           |
|---------------------------------------------------------|-----------------------------------|
| index.html / /                                          | Landing or home page              |
| register.html / /register                               | User registration                 |
| login.html / /login                                     | User login                        |
| dashboard.html                                          | Main dashboard                    |
| home_planner.html / /home-budget                        | Home planning inputs              |
| home_recommendations.html / /home-recommendations       | Home recommendations              |
| party_planner.html / /party-budget                      | Party planning inputs             |
| party_recommendations.html / /party-recommendations     | Party recommendations             |
| jewelry_planner.html / /jewelry-budget                  | Jewelry planning and image upload |
| jewelry_recommendations.html / /jewelry-recommendations | Jewelry recommendations           |
| history.html / /history                                 | Saved recommendation history      |
| /logout                                                 | Ends authenticated session        |

## 15. End-to-End Application Workflow

The workflow begins when a user registers or logs in, reaches the dashboard, selects Home, Party, or Jewelry, enters budget and domain-specific information, and submits the planner. FastAPI validates and routes the request to planner logic, which sends the context to Gemini. The backend formats the result, displays it, and stores it in History.

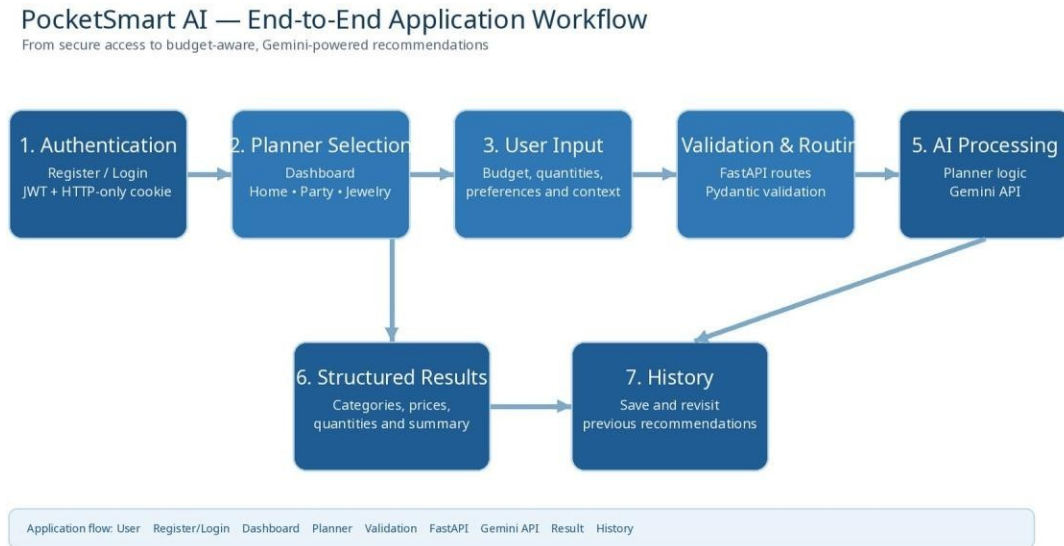


Figure 15.1 – End-to-End Application Workflow

## 16. Testing and Security

| Test Area                 | Expected Behaviour                                               |
|---------------------------|------------------------------------------------------------------|
| Registration / Login      | Valid users can create accounts and authenticate.                |
| Protected Access / Logout | Protected pages require authentication; logout ends the session. |
| Planner Inputs            | Home, Party, and Jewelry requirements are accepted.              |
| Recommendation Generation | Selected planner generates structured recommendations.           |
| Budget Summary            | Total and remaining budget are displayed.                        |
| History                   | Previous recommendation records can be viewed.                   |

Security considerations include secure password hashing, HTTP-only cookie storage for JWT/access tokens, environment-based Gemini API key protection, validation of planner inputs and uploaded images, and user-specific association of History records.

## 17. Features, Outcome and Conclusion

Single authenticated web application; Home Interior, Party, and Jewelry budget planners; Gemini API-based recommendations; budget-aware planning; category-based results; estimated prices and quantities; shopping references; optional outfit image; Recommendation History; protected routes; structured FastAPI backend; Jinja2 frontend templates.

### Project Outcome

PocketSmart AI demonstrates the complete workflow: Authenticated User Input → Planner-specific Processing → Gemini-powered Recommendation Generation → Structured Result Presentation → Recommendation History.

### Conclusion

PocketSmart AI combines budget-based planning with Generative AI to create a practical recommendation assistant. Each planner follows a common workflow from input and validation through Gemini processing, structured result presentation, and History. The modular structure can be extended with additional planning categories, richer comparisons, stronger personalization, and more detailed reporting.

## 18. Final Feature Summary and GitHub Repository

| Area               | Project Capability                                                                    |
|--------------------|---------------------------------------------------------------------------------------|
| Planning           | Home Interior, Party, and Jewelry budget planners                                     |
| AI                 | Gemini API-based personalized recommendation generation                               |
| Authentication     | Registration, password hashing, login, JWT/access token, HTTP-only cookie, and logout |
| Frontend           | Jinja2 templates with HTML/CSS                                                        |
| Recommendation     | Structured categories, prices, quantities, and suggestions                            |
| History            | Stores and displays recommendation results                                            |
| Image Support      | Optional outfit image in Jewelry workflow                                             |
| Deployment Support | Uvicorn server and environment-based configuration                                    |

The source code and project files are available in the following GitHub

repository: <https://github.com/priyadarshiniv25ds033-hash/PocketSmart-AI>

End of Project Documentation